

MAT150 | Business Statistics

Course Text

Lind, Douglas A., Marchal, William A. and Samuel A. Wathen. *Basic Statistics for Business and Economics*, 10th edition, McGraw-Hill/Irwin, 2022, ISBN: 9781260716313

The text is provided digitally as part of the course enrollment. Students may find used, new, or rental print copies by searching for the ISBN.

Required Computing Software

Several types of computer software will perform the type of statistical analyses taught in this class. For this course, the only required software is Microsoft Excel.

Course Description

This course familiarizes students with the basic concepts of business statistics and provides a comprehensive overview of its scope and limitations. Students perform statistical analyses of samples, compute the measures of location and dispersion, and interpret these measures for descriptive statistics. Other sections review linear regression, multiple regression, and correlation analysis, as well as model building, model diagnosis, and time series regression using various models. After a review of the basic concepts of probability, students apply discrete and continuous distributions of probability. Other topics include constructing a hypothesis on one and two samples, performing one-way and two-way analyses of variance, and applying nonparametric methods of statistical analysis.

Learning Outcomes

After completing this course, students will be able to:

1. Define statistics and identify its scope and limitations.
2. Describe and apply the basic concepts in statistics.
3. Apply the sampling methods and the Central Limit Theorem to perform statistical analyses of samples and to predict population behavior.
4. Compute and interpret measures of location and dispersion.
5. Represent the statistical data in different forms and interpret the different representations.
6. Perform linear regression and correlation analysis.
7. Perform multiple regression and correlation analysis.
8. Describe the basic concepts of probability.

9. Describe and apply the discrete and continuous distributions of probability.
10. Conduct hypothesis tests based on one or two samples.
11. Perform one-way and two-way analyses of variance (ANOVA).
12. Apply nonparametric methods of statistical analysis.
13. Perform model building and model diagnoses.

Course Prerequisites

Successful completion of Introductory and/or Intermediate Algebra courses is recommended before taking Business Statistics.

Academic Integrity Statement

Academic integrity is the pursuit of scholarly activity in an honest, truthful and responsible manner. Violations of academic integrity include, but are not limited to, plagiarism, cheating, fabrication and academic misconduct. Failure to comply with the Academic Integrity Policy can result in a failure and/or zero on the attempted assignment/examination, a removal from the course, disqualification to enroll in future courses, and/or revocation of an academic transcript. Penalties for academic integrity violations can be retroactively applied upon discovery, and at our discretion.

Course Completion Policy

In order for a course to be considered complete, **all required coursework must be attempted, submitted, and graded.** Required coursework consists of graded assignments. Any Academic Integrity Policy violations may prevent a course from being considered complete.

Course Evaluation Criteria

Your score provides a percentage score and letter grade for each course. A passing percentage is 70% or higher.

There are a total of 1000 points in the course:

Topic	Assessment	Points
3	Graded Exam 1	125
6	Graded Exam 2	125
6	Midterm Exam	200
8	Graded Exam 3	125
13	Graded Exam 4	125
14	Graded Final Exam	300
Total		1000

Course Roadmap *Graded items in **bold***

Intro | Introduction and Getting Started

Subtopics

- Welcome
- Academic Honesty

Assignments

- Course Introduction
- Academic Honesty and Integrity Lesson
- Statement of Academic Honesty and Integrity (required)

Topic 1 | Statistics: An Introduction and Basic Concepts

Subtopics

- Use of Statistics
- Types of Variables
- Levels of Measurement
- Ethics in Statistics
- Software and Statistics
- Graphical Displays of Categorical Data

Assignments

- Chapter 1 Reading
- Lesson 1: Statistics: An Introduction and Basic Concepts
- Assignments
- Optional Media Resources
- Topic 1 Homework

Topic 2 | Descriptive Statistics: Numerical Measures

Subtopics

- Arithmetic Mean
- Geometric Mean
- Median and Mode
- Measures of Dispersion
- Chebyshev's Theorem and the Empirical Rule
- Using Software to Compute Descriptive Statistics

Assignments

- Chapter 3 Reading
- Lesson 1: Descriptive Statistics: Numerical Measures
- Assignments
- Optional Media Resources
- Topic 2 Homework

Topic 3 | Descriptive Statistics: Representational

Subtopics

- Dot Plot, Stem Plot and Histogram
- Quartiles, Deciles, and Percentiles
- Skewness
- Bivariate Data

Assignments

- Chapter 4 Reading
- Lesson 1: Descriptive Statistics: Representational
- Assignments
- Optional Media Resources
- Topic 3 Homework

- **Graded Exam 1**

Topic 4 | Probability

Subtopics

- Probability Approaches
- Probability Calculations
- Tools of Analysis
- Computing the Number of Possible Outcomes

Assignments

- Chapter 5 Reading
- Lesson 1: Probability Distributions
- Assignments
- Optional Media Resources
- Topic 4 Homework

Topic 5 | Discrete and Continuous Probability Distributions

Subtopics

- Discrete Probability Distributions
- Binomial Probability Distributions
- Poisson Probability Distributions
- Uniform Probability Distributions
- Normal Probability Distributions
- Sampling Distribution of the Sample Mean

Assignments

- Chapter 6 Reading
- Chapter 7 Reading
- Lesson 1: Discrete and Continuous Probability Distributions
- Assignments
- Optional Media Resources
- Topic 5 Homework

Topic 6 | Sampling Methods and the Central Limit Theory

Subtopics

- Sampling a Population
- Sampling Errors
- Central Limit Theorem

Assignments

- Chapter 8 Reading
- Lesson 1: Sampling Methods and the Central Limit Theory
- Assignments
- Optional Media Resources
- Topic 6 Homework
- **Graded Exam 2**
- **Midterm Exam**

Topic 7 | Using Confidence Intervals in the Sampling Process

Subtopics

Assignments

- Large Sample Confidence Intervals
- Small Sample Confidence Intervals
- Proportions
- Sample Size

- Chapter 9 Reading
- Lesson 1: Using Confidence Intervals in the Sampling Process
- Assignments
- Optional Media Resources
- Topic 7 Homework

Topic 8 | Tests of Hypothesis

Subtopics

- Hypothesis Testing: An Introduction
- Decision Making in Hypothesis Testing
- Hypothesis Testing with Proportions
- Two-Sample Test of Hypothesis

Assignments

- Chapter 10 Reading
- Chapter 11 Reading
- Lesson 1: Tests of Hypothesis
- Assignments
- Optional Media Resources
- Topic 8 Homework
- **Graded Exam 3**

Topic 9 | Analysis of Variance

Subtopics

- Using the F Distribution in Variance Analysis
- Analysis of Variance (ANOVA)
- Computing the Analysis of Variance (ANOVA) - Sum of Squares
- Analyzing the Variance
- Use of Software in Variance Analysis

Assignments

- Chapter 12 Reading
- Lesson 1: Analysis of Variance
- Assignments
- Optional Media Resources
- Topic 9 Homework

Topic 10 | Regression Analysis

Subtopics

- Correlation Analysis
- Coefficient of Correlation
- Regression Analysis
- Confidence Interval and Prediction Intervals
- ANOVA Table

Assignments

- Chapter 13 Reading
- Lesson 1: Regression Analysis
- Assignments
- Optional Media Resources
- Topic 10 Homework

Topic 11 | Multiple Regression Analysis

Subtopics

- Multiple Regression Analysis Equation
- Analyzing ANOVA Table Output
- Analyzing Individual Independent Variables

Assignments

- Chapter 14 Reading
- Lesson 1: Multiple Regression Analysis
- Assignments
- Optional Media Resources
- Topic 11 Homework

Topic 12 | Nonparametric Methods

Subtopics

- Chi-Square Test
- Contingency Table Analysis

Assignments

- Chapter 15 Reading
- Lesson 1: Nonparametric Methods
- Assignments
- Optional Media Resources
- Topic 12 Homework

Topic 13 | Process Improvement Techniques

Subtopics

- Statistical Process Control
- Creating Control Charts
- Analyzing Control Charts
- Natural Tolerance Limits
- p Chart

Assignments

- Lesson 1: Process Improvement Techniques
- Optional Media Resources
- **Graded Exam 4**
- Preparing for Your Proctored Final Exam

Topic 14 | Review and Final Exam

Subtopics

- Complete all course assignments
- Submit the End of Course Survey

Assignments

- Lesson 1: Review
 - **Final Exam**
 - End of Course Survey
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Related Courses

**BUS106: Business
Ethics**

**BUS110: Business
Law**

**BUS120:
Organizational
Behavior**